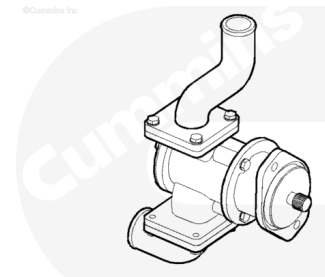


Trainings ▾

General Information

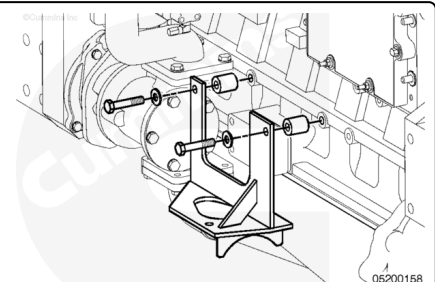
The engine uses a vane-type, positive displacement, self-priming sea water pump.



Preparatory Steps

⚠ CAUTION ⚠

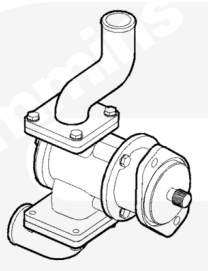
Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations



- Shut off the sea water inlet valves.
- Shut off the fuel supply and return valves (if a major kit repair is planned).
- Remove the fuel cooler (if a major kit repair is planned).
- Remove the fuel cooler mounting bracket for access to the sea water pump cover.

Remove

If a major kit repair is being performed, remove the sea water pump.



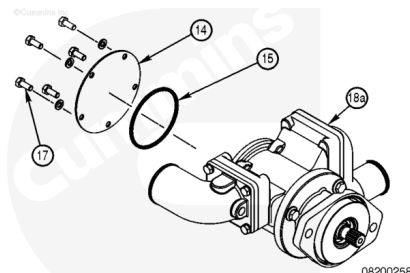
08200168

Disassemble

If a minor kit repair is being performed, remove the cover plate.

Remove the five capscrews (17) and the washers and the cover (14) from the sea water pump.

Remove and discard the quad-ring (15).



08200258

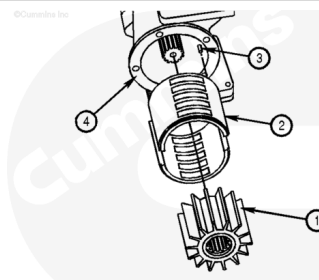
⚠ CAUTION ⚠

Do not pry against the cam liner to remove the impeller as this will break the cam liner.

Note : The two-piece cam liner will separate when removed. Do not drop the cam liner.

Check the direction of the impeller blades to make sure the impeller is properly installed at assembly.

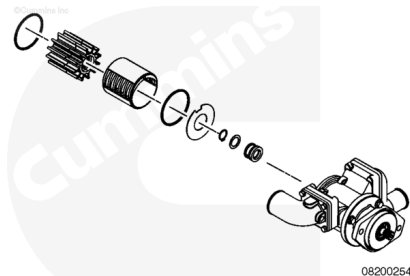
Remove the impeller (1), two-piece cam liner (2), and pin (3) from the sea water pump (4).



08200149

Remove the second quad ring and wearplate.

Remove the retaining ring from the shaft. Slide the washer and mechanical seal off of the shaft.



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Remove the bolts holding the body to the housing.

Slide the housing off of the shaft and body assembly.

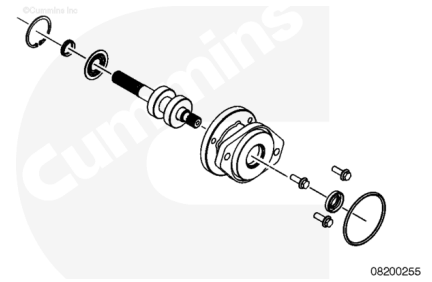
Press the seal seat out of the housing.

Remove the retaining ring from the body.

Press the bearing and shaft assembly out of the body.

Remove V-ring and bearing shield from the shaft.

Remove the lip seal.



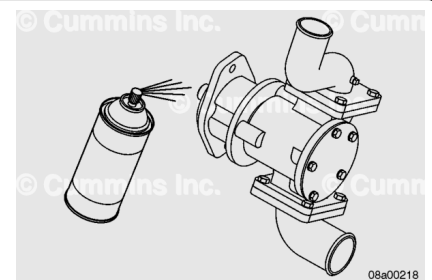
Clean and Inspect for Reuse

Clean the internal pump surfaces.

Lubricate the housing with silicone, Glycerine or a non-petroleum based lubricant.

Petroleum-based lubricant will damage the rubber impeller.

If non-petroleum-based lubricant is **not** readily available, use soapy water to ease installation.



Assemble

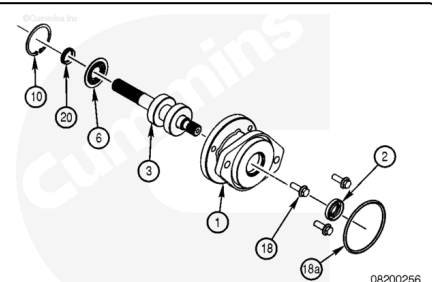
Press the lip seal (2) into the pump body (1) with the backing ring of the lip seal toward the main pump housing.

Align the lip seal so that it is uniformly pressed in the pump body. Press the shaft and bearing assembly (3) into the pump body (1).

Slide the bearing shield (6) into the pump body and against the inner bearing.

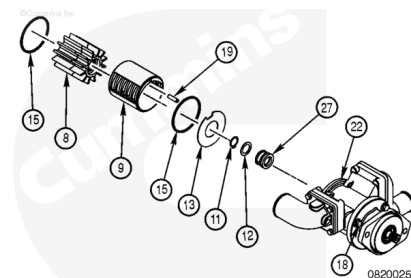
Place the V-ring (20) on the shaft against the bearing shield, with the open end of the V-ring toward the bearing assembly.

Use snap ring pliers to install the retaining ring (10) into the body.



⚠ CAUTION ⚠

Do not use petroleum based lubricant. A petroleum based lubricant will cause damage to the impeller.



Note : Grease can be used to hold the pin in place during installation.

Press the mechanical seal (27) into the housing with the white ceramic seal face toward the impeller.

Lubricate the outside of the seat boot with a water-soluble lubricant or soapy water for easier assembly.

Slide the housing (22) over the shaft and body assembly.

Install the three bolts (18).

Torque Value: 34 n•m [25 ft-lb]

Install the washer (12) onto the shaft and use snap ring pliers to install the retaining ring (11) into the groove in the shaft (3) to compress the mechanical seal.

Install the wearplate (13) with the hole in the plate lined up with the slot in the housing.

Install the quad ring (15) into the housing on top of the wearplate.

Assemble the cam halves (9) together and install the pin (19) into the cam liner.

Place the cam assembly into the housing, align the pin with the slot in the housing.

Insert the impeller (8) in the housing (22) using a twisting motion.

Make sure the impeller blades are in the same orientation as when removed.

Apply a silicon lubricant to aid in the installation of the impeller.

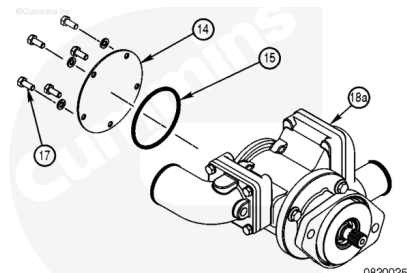
Insert the remaining quad ring (15) into the groove.

Install the cover (14) onto the housing (18a).

Install bolts (17).

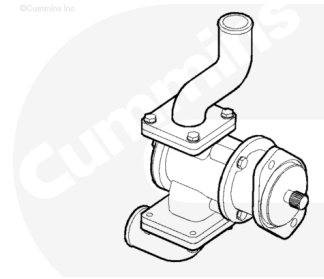
Torque Value: 11 n•m [96 in-lb]

Install the front flange sealing o-ring onto the pump body.



Install

If a major kit repair was performed, install the sea water pump.



Finishing Steps

- Install the fuel cooler mounting bracket.
- Install the fuel cooler (if a major repair kit was installed).
- Open the fuel supply and return valves (if the fuel cooler was removed).
- Open the sea water inlet valve.
- Operate the engine and check for leaks.

